

01 — Project Definition & Architecture

Turn the idea into a clearly scoped, buildable ML system

Build guide for the Context-Aware Online Distress Severity Detection project.

1. What You Are Building

Working title: Context-Aware Online Distress Severity Detection.

One-line goal: build an ethical NLP classifier that distinguishes contextual distress from ordinary negative or figurative online language and predicts an ordered severity category.

This is not a generic sentiment-analysis project and should not be presented as a clinical diagnostic system.

The core ML task is four-class severity classification: Control, Low Stress, Moderate Distress, Severe Crisis.

The output should eventually contain a predicted class, a probability distribution, and uncertainty/calibration information.

2. Why Standard Sentiment Analysis Fails

Online language contains sarcasm, slang, non-standard spelling, exaggeration, self-deprecating humor, indirect distress, and emotional ambiguity.

A keyword such as 'killing me' can describe figurative frustration rather than a genuine crisis. Conversely, genuine distress may be expressed without obvious negative keywords.

Therefore the system must model context rather than relying on word-level polarity.

3. First-Principles Architecture

The architecture is a sequence of data protection, preprocessing, representation learning, classification, uncertainty analysis, and evaluation.

The important design rule is that sensitive text is sanitized before embedding generation.

4. End-to-End Architecture

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RAW ONLINE POSTS
↓
DATA COLLECTION / DATASET IMPORT
↓
ETHICAL SANITIZATION
PII removal + de-identification
↓
DATA VALIDATION / SPLITTING
↓
DOMAIN TOKENIZATION
BPE / transformer tokenizer
↓
TRANSFORMER ENCODER
RoBERTa / domain-adapted model
↓
CLASSIFICATION HEAD
↓
4 SEVERITY PROBABILITIES
↓
CALIBRATION + UNCERTAINTY
↓
ATTENTION / ATTRIBUTION ANALYSIS
↓
METRICS + ERROR ANALYSIS
```

5. Label Design

Recommended final label hierarchy:

- Control — no meaningful distress signal in the annotation framework.

- Low Stress — ordinary or mild stress.
- Moderate Distress — substantial distress signals.
- Severe Crisis — the highest-risk linguistic category represented by the dataset.

These classes are ordinal, so the model should eventually be evaluated as more than four unrelated labels.

6. Scope Boundaries

Do not build automated diagnosis, treatment recommendations, or autonomous crisis intervention into the MVP.

The project should make clear that predictions are dataset-dependent linguistic classifications, not clinical conclusions about a person.

Do not start with a live social-media scraping system, authentication, a database, Kubernetes, or a polished frontend. Those are downstream engineering tasks.

7. Research Questions

RQ1: Do contextual transformers outperform classical NLP for this task?

RQ2: Does domain adaptation improve performance on informal online language?

RQ3: Do class weighting and focal loss improve recall for severe examples?

RQ4: Does ordinal classification better represent the severity hierarchy?

RQ5: Does probability calibration make confidence more trustworthy?

RQ6: What types of language cause systematic model failures?